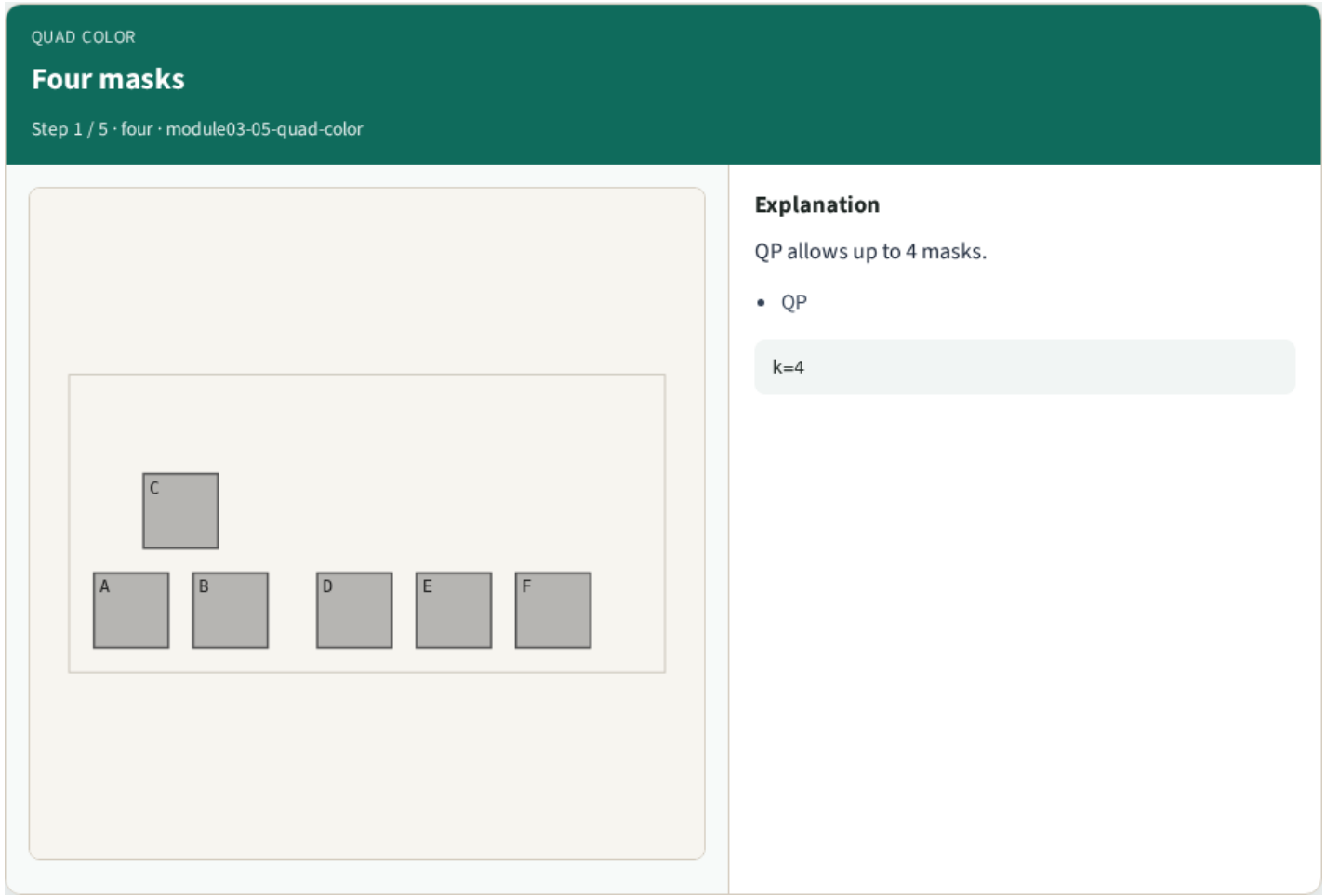
Quad patterning

Quad patterning reserves four masks; greedy coloring shows headroom when fewer suffice

The idea

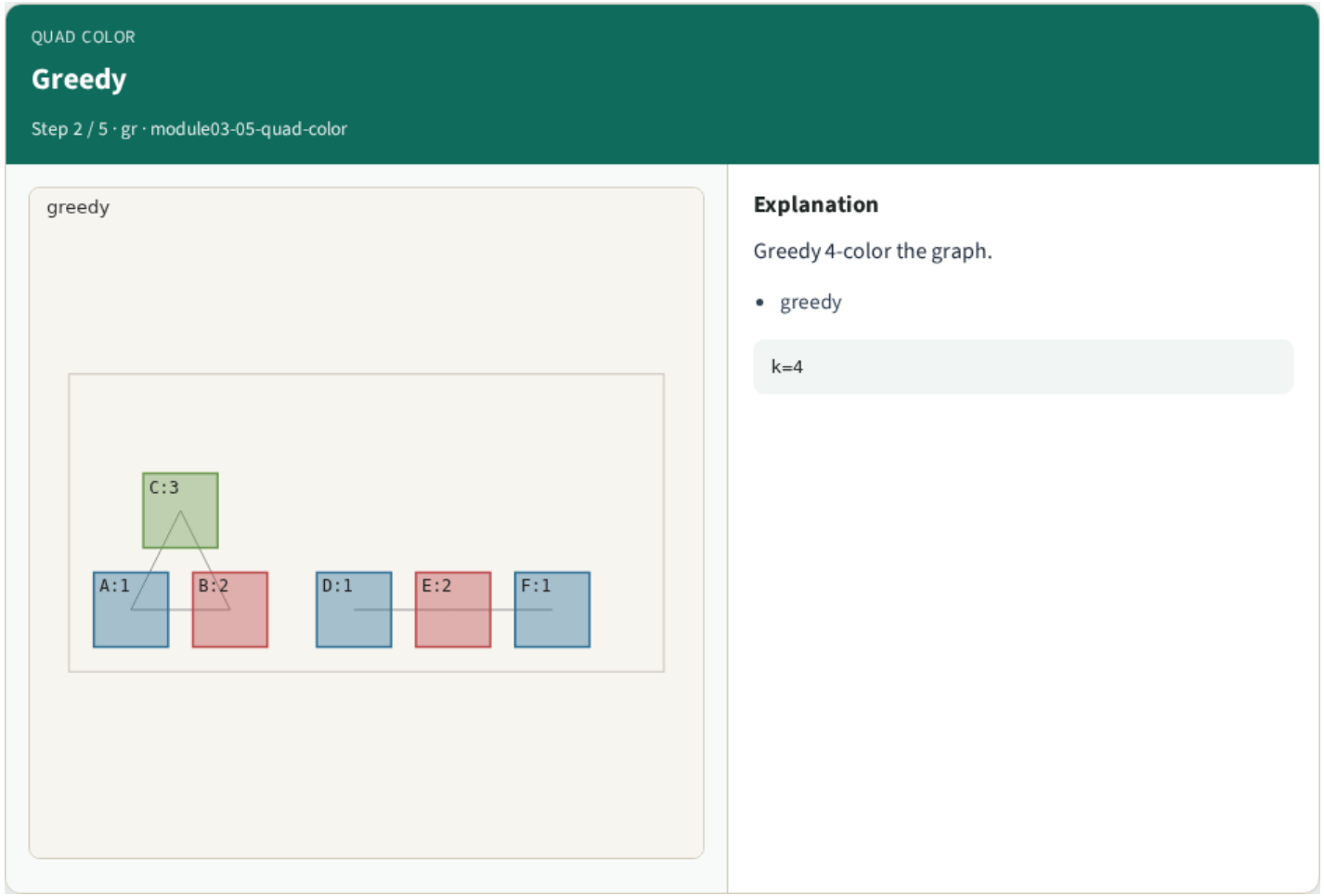
- GOLDEN: colors_used=3, k=4, ok=True on tiny_mp

Four masks



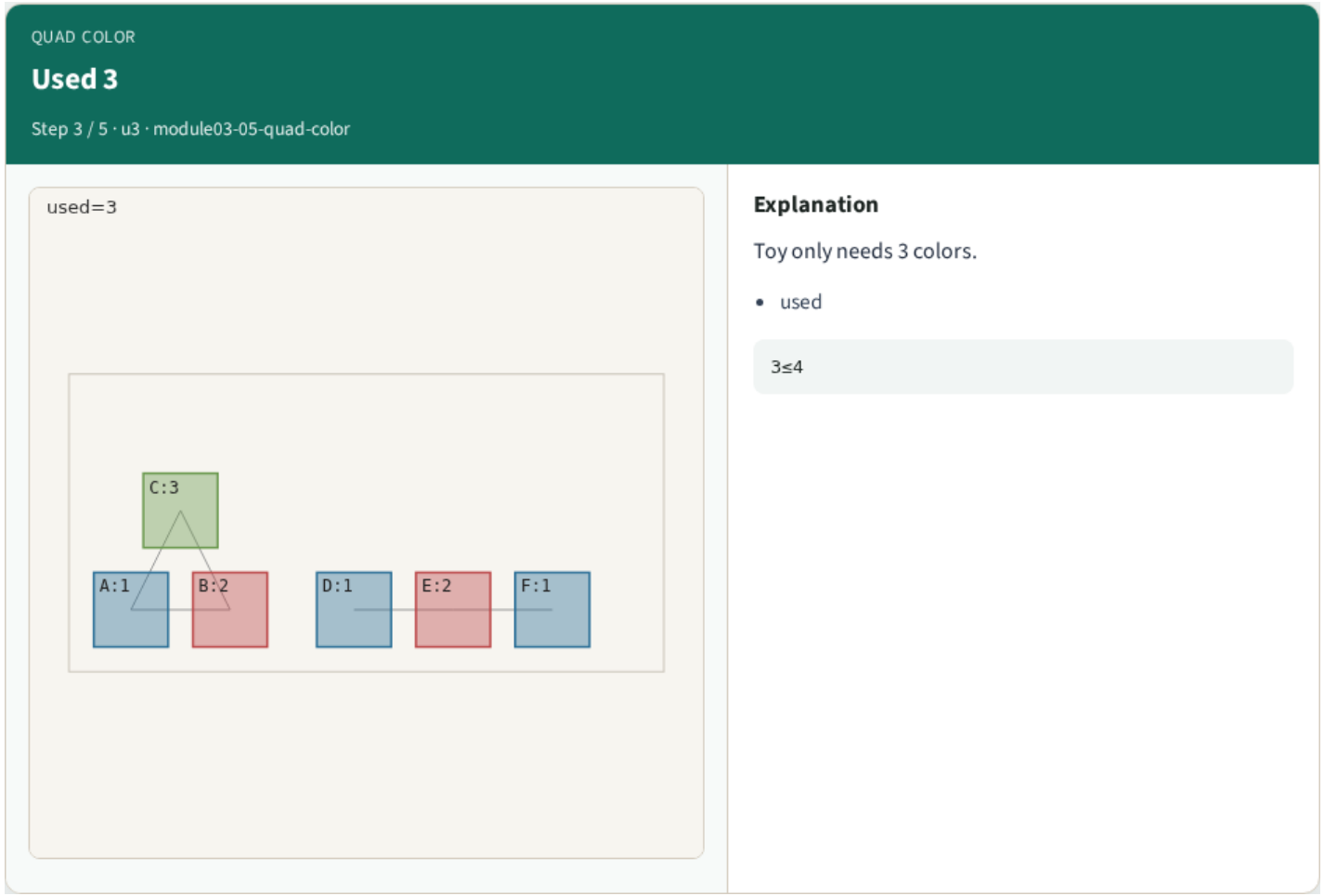
Four masks

Greedy



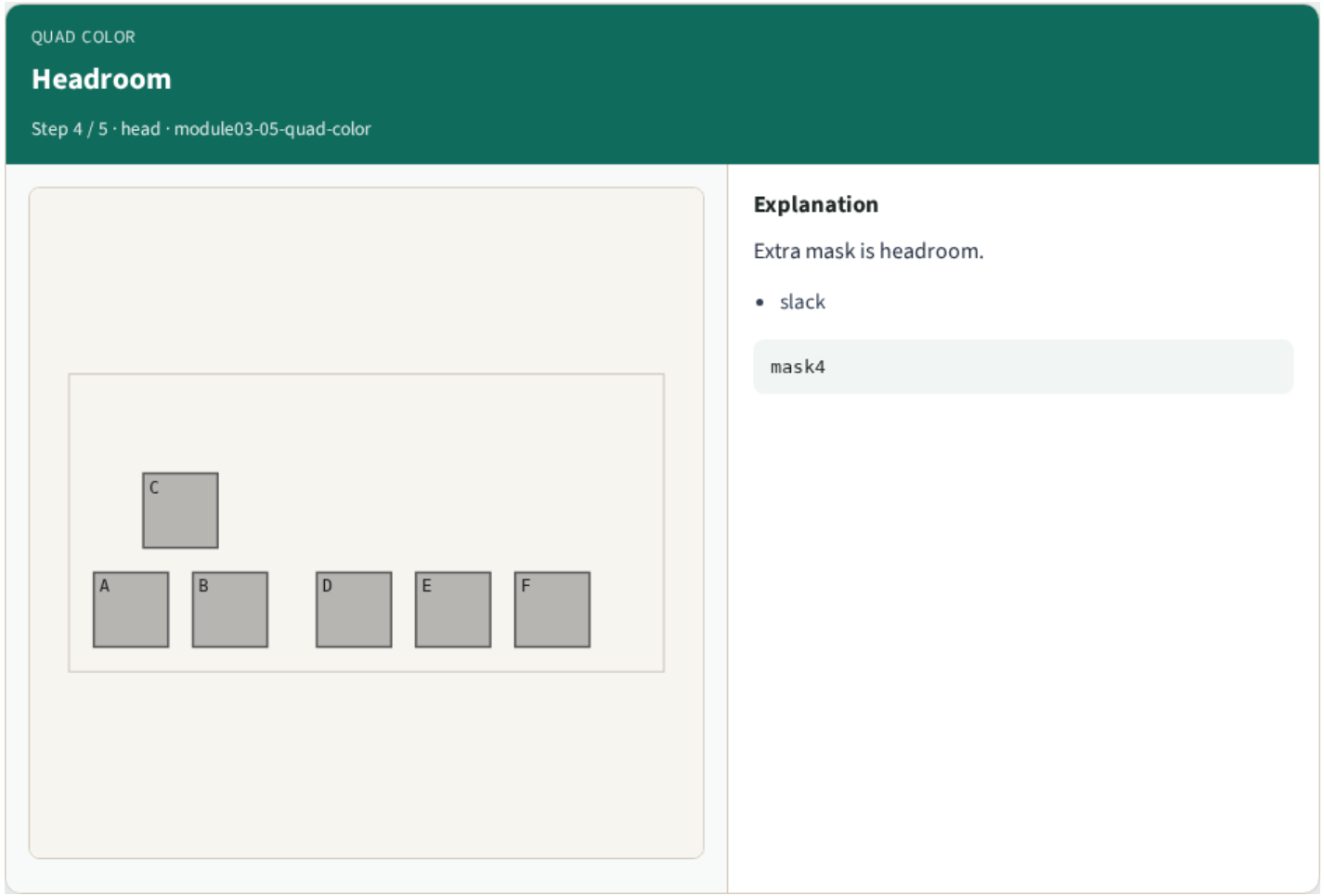
Greedy

Used 3


QUAD COLOR
Used 3
Step 3 / 5 · u3 · module03-05-quad-color
used=3
C:3
A:1
B:2
D:1
E:2
F:1
Explanation
Toy only needs 3 colors.
used
3≤4

Used 3

Headroom



Headroom

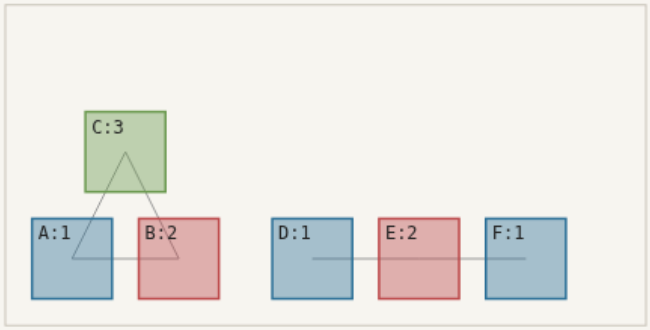
OK

QUAD COLOR

OK

Step 5 / 5 · ok · module03-05-quad-color

ok=true



Explanation

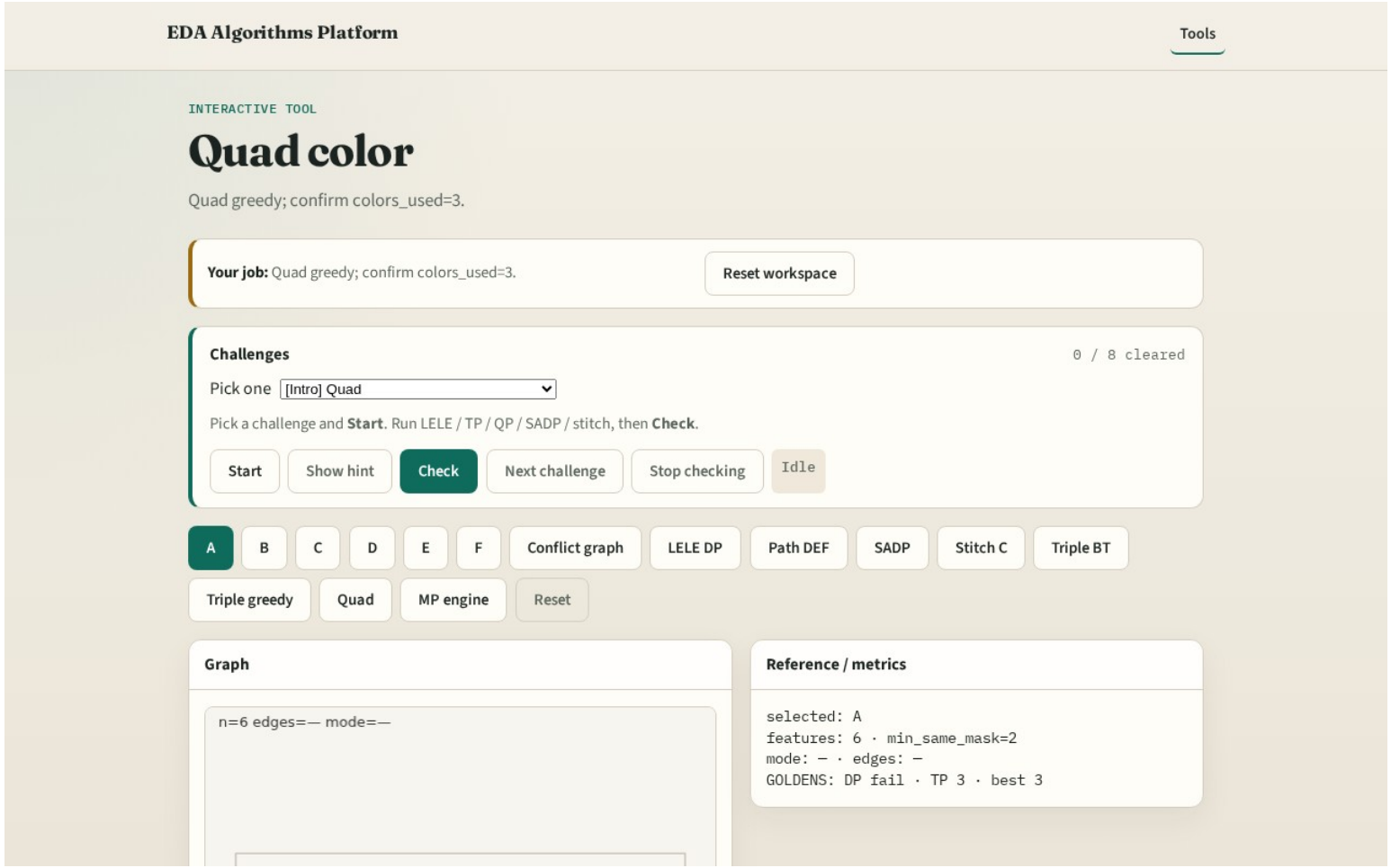
GOLDEN qp ok colors_used=3.

- GOLDEN

```
ok
```

OK

Browser lab track



Browser lab starter

Implement track

- Call quad_pattern(method='greedy'); assert ok and colors_used==3

Pitfalls

- Requiring all four masks used
- Confusing QP with TP

Your turn

- Next: multi-patterning engine pass